



Contents lists available at ScienceDirect

Combustion and Flame

journal homepage: www.elsevier.com/locate/combustflame

An experimental and modeling study on the low temperature oxidation of surrogate for JP-8 part I: Neat 1,3,5-trimethylbenzene

Bing-Yin Wang^{a,b}, Dan Yu^a, Guan-Fu Pan^{a,b}, Yue-Xi Liu^{a,b}, Jun-Jie Weng^a, Zhen-Yu Tian^{a,b,*}

^a Institute of Engineering Thermophysics, Chinese Academy of Sciences, Beijing 100190, China

^b University of Chinese Academy of Sciences, Beijing 100049, China

ARTICLE INFO

Article history:

Received 13 July 2017

Revised 26 September 2017

Accepted 23 December 2017

Available online xxx

Keywords:

1,3,5-Trimethylbenzene

Surrogate

Kerosene

Low-temperature oxidation

JSR

Reaction flux analysis

ABSTRACT

This work describes the experimental and modeling study of low temperature oxidation of 1,3,5-trimethylbenzene (T135MB) in a jet-stirred reactor over the temperature range of 700–1100 K at atmospheric pressure under fuel-lean and stoichiometric conditions. 9 C₀–C₅ hydrocarbons, 6 oxygenated products and 6 aromatic compounds were identified and quantified using GC and GC-MS. A detailed kinetic based on T135MB model of Diévert et al. was proposed to simulate the low-temperature experimental results in the present work. Rate constants of T135MB decomposition and metatheses reactions were calculated with CBS-QB3 method implemented in Gaussian 09. The performance of proposed mechanism in reproducing the experimental data is reasonably good. Reaction flux analysis shows that dominant consumption channels for T135MB oxidation are H-abstraction reactions to form 3,5-dimethylbenzyl radicals, while reactions with O/OH radicals to generate 1,3,5-trimethylphenoxy/1,3,5-trimethylphenyl and ipso-addition to form m-xylene play minor roles. Sensitivity analysis reveals that H-abstraction from side methyl groups of T135MB by OH radical is the most inhibiting reaction oxidation at $\Phi = 1.0$, while it is a promoting reaction at $\Phi = 0.4$. Moreover, current model were validated against experimental results on T135MB oxidation in flow reactor from Diévert et al. as well as global combustion property ignition delay times from Rao et al. and Diévert et al. with reasonable predictions.

© 2018 The Combustion Institute. Published by Elsevier Inc. All rights reserved.

1. Introduction

Combustion of jet fuels in engines is the main power source of civilian and military aircraft. Since jet fuels consist of multiple classes of complex components [1,2] and it is unrealistic to establish a detailed kinetic model to emulate the combustion characteristics of real fuel [1,3], it has been widely accepted to use surrogate fuel to simplify the chemical kinetic for real jet fuel [4,5]. Surrogate fuel are comprised of several components but be capable of reproducing the gas phase combustion characteristics of studied real fuel. Aromatic compounds are major components in many sorts of jet fuels, and understanding their combustion characteristics is very important in optimizing engine performance and exploring the influence of fuel variability on pollutant formation and engine efficiency. The preference of 1,3,5-trimethylbenzene (T135MB) as representative aromatic specie in practical surrogate fuel is due to the symmetry of molecule structure allowing a simplification of model construction. It is also utilized to increase the average molecular weight of surrogate fuel and provide further insight into

the combustion characteristic and soot formation study of surrogate fuel [6]. Thus, it is meaningful to establish a detailed T135MB kinetic mechanism and couple it with computational fluid dynamics.

In recent years, T135MB, one crucial component in surrogate fuel, has captured attention in the measurement of macroscopic combustion parameters [6–11]. Experimental data and chemical kinetic mechanism of T135MB have been studied mostly at relatively high temperature (>1000 K). Previous T135MB work was mainly investigated with shock tube [12], laminar [13,14] and counter-flow diffusion flames [15]. High temperature flow reactor oxidation [6,7,16], diffusion flame extinction limits [11,17,18], ignition delay times and laminar burning velocities [8,9,19,20] were carried out to study the combustion characteristic of T135MB. To facilitate the comparison between different methods, experimental conditions for previous T135MB-related studies were included in Table 1. These experimental results, including both global combustion properties and chemical details are then used to examine newly developed kinetic mechanism in current work.

Roubaud et al. [7] measured the auto-ignition feature of T135MB at stoichiometric condition with a rapid compression machine, over temperature range of 600–900 K and pressure of 5–25 bar. Rao et al. [10] studied the ignition delay times of T135MB

* Corresponding author.

E-mail address: tianzhenyu@iet.cn (Z.-Y. Tian).

<https://doi.org/10.1016/j.combustflame.2017.12.030>

0010-2180/© 2018 The Combustion Institute. Published by Elsevier Inc. All rights reserved.

Table 1

Experimental condition for T135MB studies in the present work and the literature.

Experiments	Temperature	Pressure	Φ	Ref.
Flow reactor	638–948 K	12.5 atm	1.0	[6]
Shock tube reactor	1017–1645 K	20, 50 atm	0.51–1.86	[21]
Jet-stirred reactor	700–1100 K	1 atm	0.4–1.0	Present work
Flame extinction limits	300, 500 K	1 atm	0.6–1.3	[11]
	400 K	1 atm	0.8–1.6	[8]
Laminar flame speeds	353 K	1 atm	0.7–1.5	[9]
	400 K	1 atm	0.71–1.39	[6]
	400, 700 K	1 atm	0.7–1.4	[8]
Ignition delay times	600–900 K	5–25 bar	1.0	[7]
	1080–1560 K	1.0–5.0 atm	0.5–2.0	[10]
	1111–1468 K	10, 20 atm	0.5–2.0	[6]

Table 2

Summaries of relevant T135MB kinetic model.

Model	Species	Reactions	Source of T135MB sub-model	Ref.
Present model	910	5330	calculation	Present work
Diévar et al.	450	4569	toluene	[6]
Gudiyella et al.	202	882	m-xylene	[21]

with shock tube over $\Phi = 0.5$ –2.0, 1080–1560 K and 1.0–20.0 atm. Won et al. [11] found that extinction limits of T135MB/air diffusion flames were determined by the capability of fuel to establish a radical pool governed by molecular structure. Hui et al. [8] and Ji et al. [9] measured the premixed extinction limits of T135MB/O₂/N₂ mixtures ($\Phi = 0.8$ –1.6 at 400 K and 700 K) and the laminar burning velocities of premixed T135MB/air flames (counter-flow configuration, under atmospheric pressure and 353 K), respectively. Compared to the macroscopic combustion study, speciation investigation in the oxidation of T135MB are relatively limited. The speciation profiles for T135MB oxidation at 1017–1646 K, equivalence ratios from 0.51 to 1.86 and pressure of 20 and 50 atm in a reflected shock tube were reported by Gudiyella and Brezinsky [21]. Recently Diévar et al. [6] presented the reflected shock ignition delay times, laminar burning velocities and oxidative reactivity profiles in a high-pressure flow reactor at 12.5 atm, showing that neat T135MB has no negative temperature coefficient (NTC) effect. An estimated kinetic model proposed to predict the species profiles can simulate some specific experimental results for high temperature chemistry of T135MB. However, it has not been extensively validated. There is even scarce work on the low-temperature oxidation of T135MB in micro-kinetic perspective regarding the speciation analysis and observation of more stable intermediates species. Low-temperature kinetic is very crucial to reveal the ignition process and validate the universality of proposed models. Thus, it is desired to study the low temperature oxidation of T135MB. Relevant T135MB kinetic models were summarized in Table 2.

In this work, low temperature oxidation of T135MB within jet-stirred reactor (JSR) system was investigated over temperature range of 700–1100 K and equivalence ratios of 0.4 and 1.0. The species measurements were conducted by gas chromatographic techniques. Goal of present study is to develop a more extensively universal kinetic mechanism for combustion of T135MB. Theoretical calculations have been performed to acquire more appropriate rate constants for key elementary steps and low-temperature kinetic characteristics of T135MB oxidation was performed using the current updated model. These results will provide more insights into better understanding the low temperature oxidation of T135MB and formation of stable intermediates.

2. Experimental

The detailed information of current low-temperature oxidation system was described in earlier publications [22,23], therefore only

a brief introduction is given here. A spherical fused silica JSR with 50 mm inner diameter was designed for low-temperature oxidation of T135MB from 700 to 1100 K at atmospheric pressure. The whole system consists of high pressure infusion pump (FL2200, China), evaporation chamber (self-assembled, diameter of 5 cm, able to be heated above 600 K), JSR, GC (Agilent 7890B, USA) and GC-MS (Agilent 7890B-5977A, USA), as shown in Fig. S1 in the Supplemental Material (SM). Residence time τ is calculated by equation $\tau = V/Q$. V and Q stand for volume of used JSR and volume flow rate of mixture in JSR. Experimental conditions are listed in Table 3. Liquid T135MB with feeding rate of 0.061 ml/min (1% in gas phase) was heated to 473 K above boiling point (438 K) and carried into JSR by Ar (purity of 99.999%). Flow rate of oxygen (purity of 99.999%), controlled by MKS mass-flow controller, is 30% and 12% of total 1.000 standard liter per minute diluted in Ar, corresponding to $\Phi = 0.4$ and 1.0, respectively. All transfer lines were preheated to 473 K to eliminate temperature gradients and experimental uncertainties, particularly for aromatic intermediates with high boiling point. Target temperature was achieved by a regulated heating controller (HT60, Horst Germany) and measured by a K-type thermocouple located inside the center space of sphere. The outlets were analyzed by online GC using thermal conductivity detector (TCD) and flame ionization detector (FID), which is equipped with capillary columns (Al₂O₃-KCl and HP-INNOWax). Good experimental reproducibility and repeatability were guaranteed by multiple measurements at each experimental condition. Uncertainty of temperature measurement was estimated to be within ± 5 K. Estimated experimental uncertainties of mole fractions were within $\pm 5\%$ for major species and $\pm 10\%$ for intermediates.

3. Modeling

A detailed kinetic model of T135MB oxidation was developed based on mechanism proposed by Diévar et al. [6]. The detailed kinetic model includes a C₀–C₂ core subset, a mechanism to simulate the oxidation of C₃–C₅ and aromatic species, especially T135MB sub-mechanism. Rate constants of some reactions in T135MB sub-mechanism proposed by Diévar et al. [6] were estimated by analogy to that of toluene. Sensitivity analysis under current experimental conditions revealed that unimolecular decomposition and H-abstraction reaction are two types of key pathways in fuel destruction. It should be noted that rate constants of T135MB decomposition and H-abstraction were simply considered three times as those reactions of toluene in the literature [6]. However, there is only one benzyl type methyl in the structure of toluene,

Table 3
Experimental conditions.

Φ	C/O	X_{T135MB}	X_{O_2}	X_{Ar}	TFR (sccm)	BP (K)	τ (S)	P (atm)
0.4	0.150	1.00	30.00	69.00	1000.00	438	1.06–1.67	1
1.0	0.375	1.00	12.00	87.00	1000.00	438	1.06–1.67	1

Note: X_i is the inlet percentage of species i ; TFR, BP and τ refer to the total flow rate, boiling point and residence time, respectively.

while T135MB molecular features three benzyl methyl groups with large steric hindrance and inductive effect. The steric hindrance effect will slow reaction rate constant. Oppositely, methyl group has a weak inductive effect accelerating the process of elementary reaction. Therefore, it is not appropriate to assign simply crucial elementary reactions with three times of rate constants of toluene to T135MB sub-mechanism, which would lead to substantial uncertainties.

As a result, rate constants of T135MB unimolecular decomposition and H-abstraction were calculated in this work. Detailed chemical kinetic mechanism describing the combustion property of T135MB is updated by combination new calculation and analogy results, based on experimental observations and previously proposed T135MB model [6]. The detailed construction process of T135MB sub-mechanism is described below.

Quantum chemistry calculations were performed at the CBS-QB3 level of theory [24] implemented in Gaussian 09 [25]. Rate constants of T135MB unimolecular decomposition and H-abstraction reactions were calculated by combining electronic structure theory with transition state theory. Geometry optimization and vibration frequencies analysis were performed with CBS-QB3 method. In addition, intrinsic reaction coordinate calculations were used to validate the connections between target transition states and reactants or intermediates [26]. Kinetic parameters were gained from transition state theory using ChemRate program [27] with a modified Arrhenius form.

The sub-mechanism of T135MB was mainly taken from model of Diévar et al. [6]. The kinetic model of T135MB developed by Diévar et al. [6] can reproduce the global combustion behavior of ignition delay time, laminar flame speed, diffusive and premixed strained extinction limits and flow reactor reactivity, but it was not well validated against the results of low-temperature oxidation, especially for significant intermediates. It is necessary to update the rate constants of T135MB sub-mechanism to improve the comprehensive prediction capability. Arrhenius parameters of reactions in T135MB sub-mechanisms were modified, as shown in Table 4 along with calculated and reevaluated rate constants for reactions R1–R10. Figure S2 in SM shows comparisons of reaction rate constants of calculated unimolecular decomposition and H-abstraction elementary steps with those analogous ones in model of Diévar et al. [6]. In general, calculated rate constants are slower than that of estimated ones in former T135MB sub-mechanism. Due to concerted efforts of strong steric hindrance effect and weak inductive effect of other two benzyl methyl groups in T135MB molecular structure, the actual values of unimolecular decomposition and H-abstraction of T135MB are three times slower than those of toluene. Therefore, it is believable that present T135MB sub-mechanism adopting calculated results using quantum chemistry method has a better performance in predicting the low-temperature oxidation of T135MB. The modified T135MB subset was validated against experimental results in jet-stirred reactor, speciation data obtained in a flow reactor and ignition delay times over temperature range of 600–1560 K, pressure ranging from 1 to 25 atm and equivalence ratios of 0.4–2.0.

The chemical structures and species names of some key intermediates are listed in Table S1 in SM. Current T135MB model is available in CHEMKIN format in SM. Simulation of T135MB low-

temperature oxidation was performed with PSR code in Chemkin-II software [28]. Moreover, ignition delay times of T135MB measured at different pressures were simulated using homogeneous constant volume model with SENKIN code. In order to improve the predictive capability of sub-mechanisms and figure out the key reactions, sensitivity and reaction flux analysis were performed.

4. Results and discussion

4.1. General phenomena

In the low-temperature oxidation of T135MB, 22 species were identified by continuous GC techniques. Although there are still some differences between experimental data and predicted results, they are generally acceptable. The comparisons between measured and simulated mole fraction profiles reflect that current model shows satisfactory performance in capturing the reaction temperatures and peak mole fractions for most species. Performance of kinetic model was examined with experimental results from low to high temperature aspects in Table 1 under various conditions. Sensitivity and reaction pathways analysis were performed with the current model to interpret the chemical details for T135MB low-temperature oxidation and provide further guidance for experiments and modeling study. Discussion is given in several parts with respect to different measurements. The emphasis is concentrated on the investigations of low-temperature oxidation from the present work. Information about certain crucial intermediates can facilitate understanding the combustion process and optimizing the kinetic parameters of crucial intermediate reactions. Moreover, ignition delay times from the literature are examined by the current model, which is an important macroscopical combustion property of T135MB especially at low-temperature condition.

4.2. T135MB low-temperature oxidation in JSR

Experimental and modeled mole fraction profiles of T135MB, major products (CO and CO₂) and selected minor intermediates are compared in Figs. 1–4. Comparison results of T135MB, CO, CO₂ and H₂ are shown in Fig. 1. The conversion profiles of reactants at two equivalence ratios are well predicted by the current model, as well as the formation of CO, CO₂ and H₂. The simulated mole fraction profiles of T135MB and major products by Diévar model and the present model are compared in Fig. S3 in SM. Both models can capture effectively the trends of fuel and major products within experimental uncertainties under both conditions. It is observed that the present model behaves better in reproducing the experimental results. By comparing the difference of main consumption pathways between present model and Diévar model, under-prediction of T135MB by Diévar model can be attributed to that concerted efforts of steric hindrance effect and inductive effect of methyl groups on rate constants are not considered. At low temperatures, role of HO₂ radical is very important in determining the rate of oxidation reactions. First, H-abstraction of T135MB with HO₂ radical (R5) is one of the major consumption pathways for low-temperature oxidation of T135MB. What's more, the initial stage involving HO₂ radical is the rate determining step for low-temperature oxidation of T135MB according to

Table 4

Original rate constants of the T135MB subset, rate constants are given as $k = AT^n \exp(-E_a/RT)$, units are s^{-1} , cm^3 and cal/mol . Reaction numbers in this table are not equivalent to those in the model.

	Reaction of T135MB	A	n	E_a	Log (k_1)	Log (k_2)	Ref.
R1	T135MB $\rightleftharpoons$ H + D35MB1CH ₂	1.066E10	1.694	83.6559E3	-1.13E+01	-1.44E+00	a
R2	T135MB $\rightleftharpoons$ CH ₃ + D13MBJ	1.444E25	-2.376	101101.2	-1.32E+01	-2.15E+00	a
R3	T135MB + H $\rightleftharpoons$ D35MB1CH ₂ + H ₂	1.941E01	3.980	3384.0	1.16E+01	1.27E+01	a
R4	T135MB + CH ₃ $\rightleftharpoons$ D35MB1CH ₂ + CH ₄	8.820E01	0.000	9245.0	9.06E+00	1.01E+01	a
R5	T135MB + HO ₂ $\rightleftharpoons$ D35MB1CH ₂ + H ₂ O ₂	1.191E12	0.000	14069.4	7.68E+00	9.28E+00	a
R6	T135MB + OH $\rightleftharpoons$ D35MB1CH ₂ + H ₂ O	5.310E05	2.390	-602.0	1.27E+01	1.31E+01	a
R7	T135MB + H $\rightleftharpoons$ CH ₃ + D13MB	2.848E6	2.000	0.944E3	1.18E+01	1.23E+01	a
R8	D35MB1CH ₂ + HO ₂ $\rightleftharpoons$ D35MB1CH ₂ O + OH	1.000E13	0.000	0.0	1.30E+01	1.30E+01	a
R9	D35MB1CH ₂ + CH ₃ $\rightleftharpoons$ D35MBC ₂ H ₅	1.190E13	0.000	0.221E3	1.30E+01	1.30E+01	a
R10	D13MBOJ + H(+M) $\rightleftharpoons$ D13MBOH(+M)	1.000E14	0.000	0.0	1.40E+01	1.40E+01	a
Reaction rate constants calculated and updated in T135MB sub-mechanism.							
R1	T135MB $\rightleftharpoons$ H + D35MB1CH ₂	1.3619E01	3.9299E+00	8.6676E+04	-1.47E+01	-4.14E+00	Calculated
R2	T135MB $\rightleftharpoons$ CH ₃ + D13MBJ	2.1428E01	4.2780E+00	1.0700E+05	-1.99E+01	-6.92E+00	Calculated
R3	T135MB + H $\rightleftharpoons$ D35MB1CH ₂ + H ₂	4.8944+03	3.4579E+00	6.5047E+03	8.28E+00	9.70E+00	Calculated
R4	T135MB + CH ₃ $\rightleftharpoons$ D35MB1CH ₂ + CH ₄	1.3193E-03	4.9329E+00	8.0865E+03	8.63E+00	1.05E+01	Calculated
R5	T135MB + HO ₂ $\rightleftharpoons$ D35MB1CH ₂ + H ₂ O ₂	3.7504E-04	5.1233E+00	9.8791E+03	8.07E+00	1.02E+01	Calculated
R6	T135MB + OH $\rightleftharpoons$ D35MB1CH ₂ + H ₂ O	1.3217E+03	3.1589E+00	-8.14E+02	1.24E+01	1.29E+01	Calculated
R7	T135MB + H $\rightleftharpoons$ CH ₃ + D13MB	1.71E9	1.43	5.65E3	1.15E+01	1.25E+01	b
R8	D35MB1CH ₂ + HO ₂ $\rightleftharpoons$ D35MB1CH ₂ O + OH	3.57E9	1.03	-2250.0	1.32E+01	1.31E+01	c
R9	D35MB1CH ₂ + CH ₃ $\rightleftharpoons$ D35MBC ₂ H ₅	6.12E16	-0.970	0.62E3	1.38E+01	1.37E+01	d
R10	D13MBOJ + H(+M) $\rightleftharpoons$ D13MBOH(+M)	2.5E14	0.0	0.0	1.44E+01	1.44E+01	e

Note: k_1 and k_2 are the rate constants at 700 and 1100 K; ^a referred to rate constants from Oehlschlaeger et al. [29]; ^b referred to C₆H₅CH₃ + H = C₆H₆ + CH₃ from Tian et al. [30], a times 3 to match the number of available positions; ^c referred to C₆H₅CH₂ + HO₂ = C₆H₅CH₂O + OH from da Silva et al. [31], A multiplied by 3 times; ^d Referred to C₆H₅CH₂ + CH₃ (+M) = C₆H₅C₂H₅ (+M) from Yuan et al. [32]; ^e Referred to C₆H₅O + H(+M) = C₆H₅OH(+M) from Frenklach et al. [33].

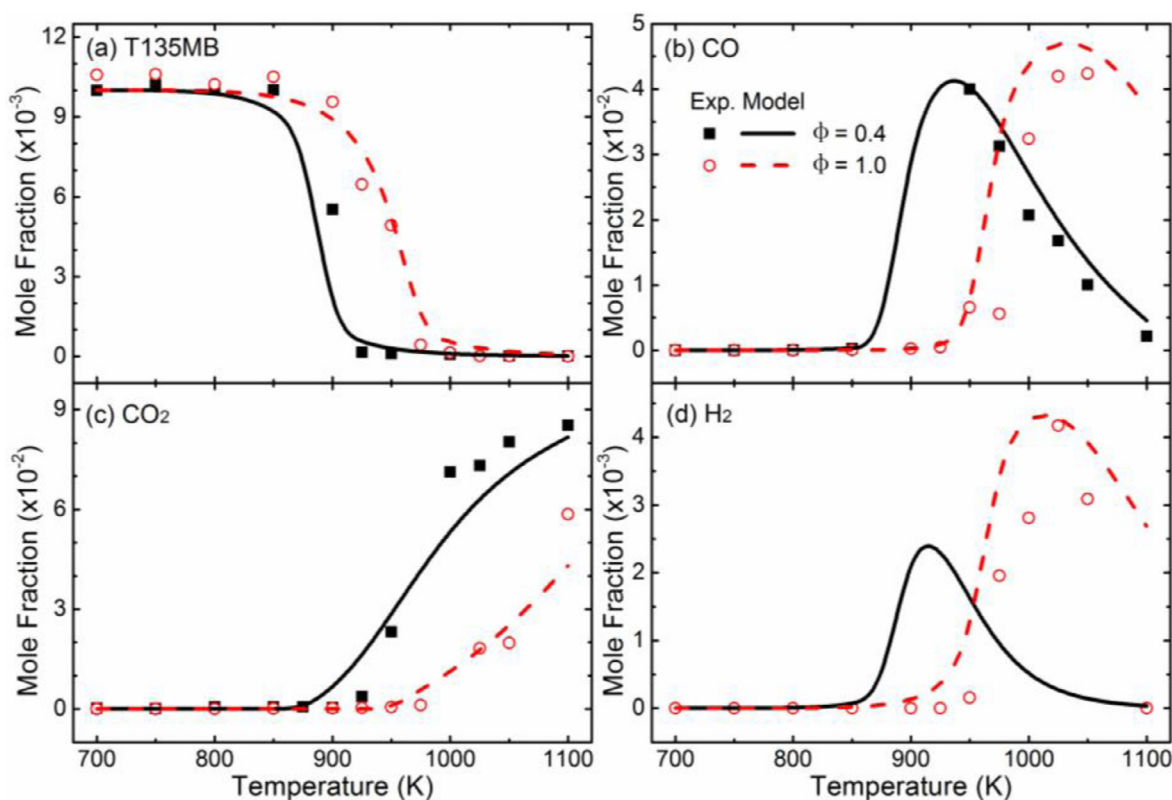


Fig. 1. Mole fraction profiles of T135MB and main products. Symbols and lines represent the experimental and simulated results, respectively.

reaction flux analysis. Reactions of HO₂ radicals with T135MB (R5), 3,5-dimethylbenzyl and other derived radicals provide main and reliable sources of OH and H radicals, determining the initial yields of OH and H radicals, whose H-abstraction reactions with T135MB (R5) are dominant consumption pathways for low-temperature oxidation of T135MB. Reaction of HO₂ radicals with T135MB (R5) is kinetically important at low temperature because of consumption

of T135MB and formation of the very reactive OH and H radicals in the reaction sequences.

As shown in Fig. S3 (a–d, f), the calculated rate constants of R1–R4, R6 are slower than those in the estimated kinetic model at low temperature. Value of rate determining step R5 is slightly faster than that of Diévar model. The concerted efforts of calculated reactions result in the onset oxidation of fuel and

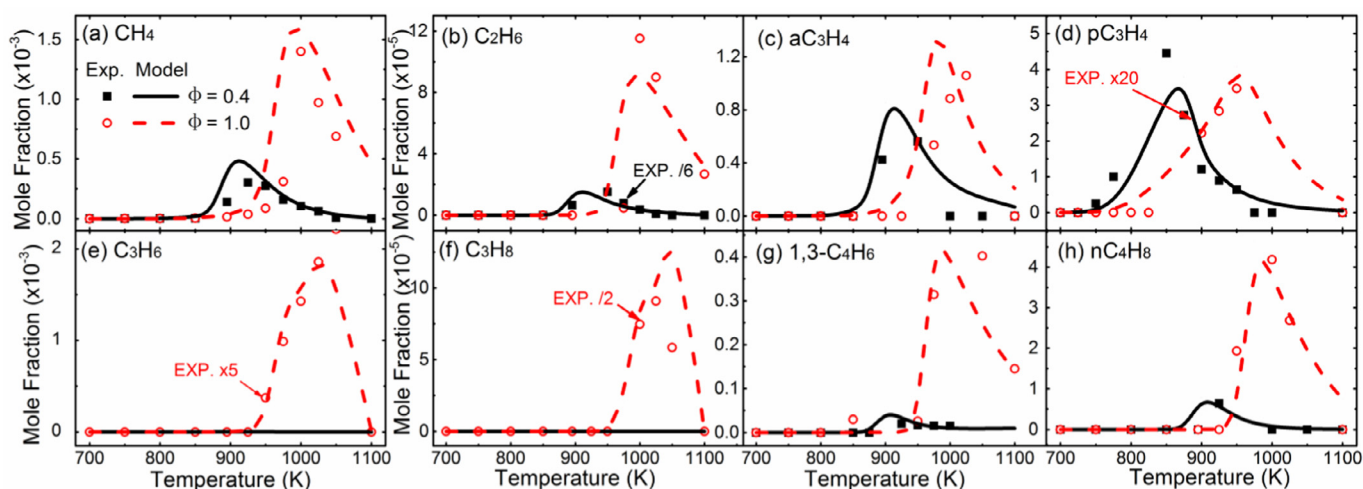


Fig. 2. Mole fraction profiles of the light hydrocarbons intermediates. Symbols and lines represent the experimental and simulated results, respectively.

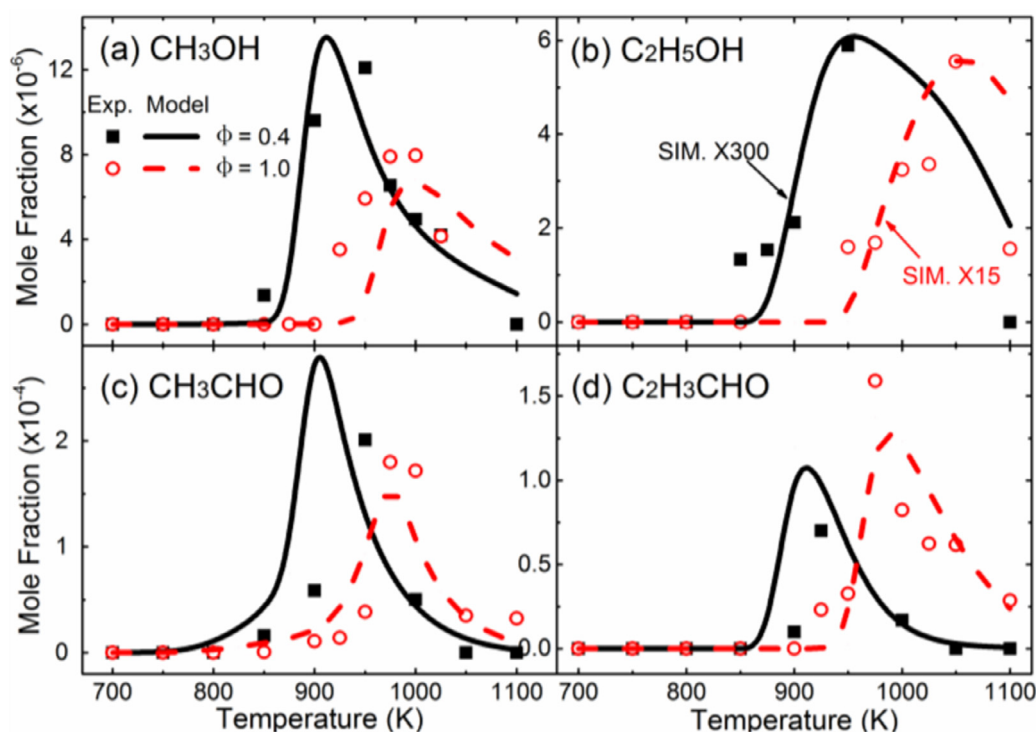


Fig. 3. Mole fraction profiles of the oxygenated intermediates. Symbols and lines represent the experimental and simulated results, respectively.

formation of products shifting to lower temperature region, which are consistent with experimental data. In both lean and stoichiometric cases, CO and CO₂ were abundantly formed as the major species. The maximum values of different species under lean condition mostly occur at around 900 K, which is lower than that temperature of 1000 K under stoichiometric condition. Under lean conditions, the maximum mole fraction of CO₂ equals to nearly theoretical production amount 9% when T135MB is completely consumed at 1100 K. The initial temperature of T135MB oxidation and formation peak of CO at $\Phi=0.4$ were both lower than those at $\Phi=1.0$. Reason for those phenomena can be attributed to the variation of reaction pathways with temperature and equivalence ratio. Under lean condition and low temperature when CO starts to form, $\text{HCO} + \text{O}_2 = \text{CO} + \text{HO}_2$, gives rise to most of CO, while as temperature increases, contribution of HCO decreases and that of another reaction: $\text{HCCO} + \text{O}_2 = \text{OH} + 2\text{CO}$ increases. As chain propagation reactions rely on the radicals provided by chain initiations,

such reaction could lead to fast decomposition of T135MB and CO formation under lean conditions. Mole fraction of CO tends to decrease accompanying with CO₂ increasing from initial oxidation of T135MB to end at both equivalence ratios. Based on ROP analysis, the dominant formation pathway for CO₂ is quite similar at both conditions, namely the reaction: $\text{CO} + \text{OH} = \text{CO}_2 + \text{H}$.

The experimental and simulated mole fraction profiles of modeling results of selected light hydrocarbon and oxygenated species are illustrated in Figs. 2 and 3. The peak temperatures of these species at $\Phi=0.4$ are almost located at 900 K, which are lower than that at $\Phi=1.0$ by 100 K. The predicted results of methane, allene, 1,3-butadiene and n-butene agree well with experimental data while those of other light hydrocarbons exhibit some discrepancies with experimental data. As one of abundant light hydrocarbons, concentration level of methane increases as equivalence ratio increases. The main formation pathways of methane change as equivalence ratio increases. Under lean condition, metathesis

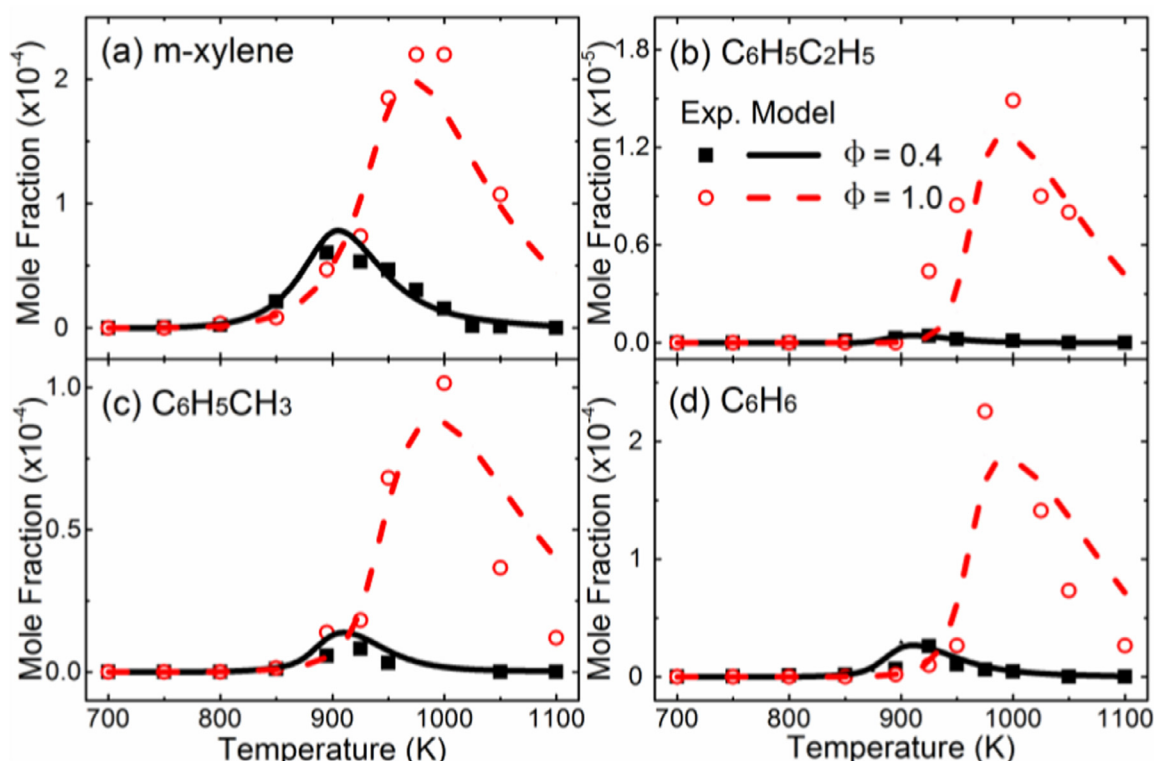


Fig. 4. Mole fraction profiles of aromatic species. Symbols and lines represent the experimental and simulated results, respectively.

reaction $\text{T135MB} + \text{CH}_3 = \text{D35MB1CH}_2 + \text{CH}_4$ and dehydrogenation $\text{CH}_3 + \text{HO}_2 = \text{CH}_4 + \text{O}_2$ account for two dominant formation routes of methane with contribution of 26% and 22%, respectively. As equivalence ratio increases, dehydrogenation of T135MB tends to contribute more (47%), while the latter becomes less dominant with a contribution of 2%. It can be seen that peak concentration of methane under lean condition is about 4 times larger than that of methane under stoichiometric condition due to the variation of formation paths. Moreover, the remaining light hydrocarbons have similar trends of methane on the formation temperature and peak concentration as a result of equivalence ratio. Mole fractions of propyne and propene are over-predicted by current model under stoichiometric condition. According to ROP analysis, formation of propyne and propene are relevant to a series of decomposition reactions of aromatic intermediates including 3,5-dimethylphenyl radicals, which are easy to be condensed during the sampling process due to the property of high boiling point. Although the whole system was heated by heating belt, quantifications experimentally of aromatic intermediates were still lower than actual value. The more steps the formation of aromatic intermediates involved in, the larger experimental uncertainties will be. On the opposite, the chain propagation of formation of light saturated hydrocarbon intermediates like ethane and propane is shorter than that of unsaturated ones. According to ROP analysis, the above two saturated intermediates are produced by the addition of methyl groups, which are formed at the initiation of oxidation of T135MB. Under-prediction of methyl radical results in the underestimation of ethane and propane.

The oxygenated intermediates show an opposite distribution rule compared to the hydrocarbons. On the whole, the peak temperatures of oxygenated species measured at $\Phi = 0.4$ are lower by about 100 K than those at $\Phi = 1.0$. ROP analysis shows that ipso-substitution of small hydrocarbon intermediates by OH and O_2 dominate the formation of oxygenated intermediates. Due to relative higher concentration of O_2 and OH under lean condition,

peak values of mole fraction of oxygenated species under fuel-lean condition are generally larger than that under stoichiometric condition, except for acrolein. Methane was mainly formed by two reaction pathways: the addition of methyl radical with molecular oxygen and consequent reaction with OH: $\text{CH}_3 + \text{O}_2 = \text{CH}_3\text{O}_2$ and $\text{CH}_3\text{O}_2 + \text{OH} = \text{CH}_3\text{OH} + \text{O}_2$, respectively. Most of acetaldehyde comes from the metathesis of propenyl radical with oxygen molecular. It is observed that current model can reproduce the measured methane and acetaldehyde profiles. However, mole fraction of ethanol is largely underestimated by current model. This phenomenon can be explained by the reaction pathway of ethanol formation. ROP analysis reveals that the addition of ethyl and OH radical has more than 80% contribution to the formation of ethanol. Underestimation of ethane causes the largest discrepancies between experimental data and simulated results. As an important intermediate during the low-temperature oxidation of T135MB, acrolein is dominantly produced from the metatheses of 1,3-butadienyl through reaction: $\text{C}_4\text{H}_5\text{-N} + \text{O}_2 = \text{C}_2\text{H}_3\text{CHO} + \text{HCO}$ under both conditions. This causes that the trends of speciation profiles of acrolein are similar to that of 1,3-butadiene under two equivalence ratios, which is the main source of 1,3-butadienyl radical.

Figure 4 compares the experimental and modeling results of selected aromatic stable intermediates at two equivalence ratios, which show good agreement between experiments and current T135MB model predictions. Aromatic intermediates are significant in the combustion of T135MB, especially in low-temperature oxidation of T135MB, due to the crucial role in both fuel oxidation and PAHs formation. M-xylene is a significant primary low-temperature oxidation product of T135MB. ROP analysis shows that ipso-substitution of T135MB by H dominates the formation of m-xylene under both equivalence ratios (75% and 81%). The predictions of ethylbenzene are slightly lower than their experimental results. These underpredictions can be accounted for that almost of ethylbenzene come from the reac-

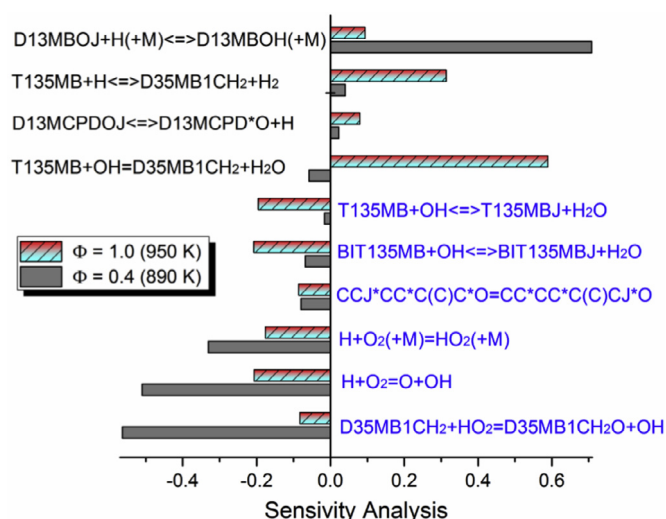


Fig. 5. Sensitivity analysis of T135MB oxidation under lean and stoichiometric conditions, corresponding to 890 K (70% conversion) and 950 K (70% conversion), respectively.

tion sequence: $C_6H_5CH_3 \rightarrow C_6H_5CH_2 \rightarrow C_6H_5C_2H_5$. And the under-prediction of toluene results in the underestimation of ethylbenzene. Due to the easy production and high concentration levels of toluene under stoichiometric condition, this reaction sequence produces larger concentration of ethylbenzene than that under lean condition. Experimental peak temperature of toluene at $\Phi = 0.4$ is obviously lower than that at $\Phi = 1.0$. By analyzing the reaction pathway of toluene production at two equivalence ratios, main production pathway of toluene is dehydrogenation reaction of 1,3-dimethyl cyclopentadiene. Difference of peak concentration of toluene between lean and stoichiometric condition is caused from different formation pathways. Under lean condition, toluene is mainly produced by ipso-substitution of m-xylene with H atom and partly by decomposition of 1,3-dimethyl cyclopentadienyl radical. While under stoichiometric condition, more than 80% of toluene is produced by the decomposition of 1,3-dimethyl cyclopentadienyl radical. Reaction rate of this decomposition step at $\Phi = 1.0$ is much higher than that at $\Phi = 0.4$ which leads a higher production of toluene under stoichiometric condition.

Sensitivity analysis for consumption of T135MB at two equivalence ratios at 70% conversion is shown in Fig. 5. Sensitivity analysis indicates that the calculated H-abstraction with HO_2 to form 3,5-dimethylbenzyl radical is the most promoting reaction for oxidation of T135MB at $\Phi = 0.4$. Meanwhile, the dehydrogenation reactions of T135MB and BIT135MB with OH radical to form 1,3,5-trimethylphenyl are the most promoting reaction for oxidation of T135MB at $\Phi = 1.0$. H-abstraction from side methyl groups by OH radical is the most negative reaction under stoichiometric condition, while this reaction plays a less promoting influence under fuel-lean condition. Hydrogenation of 1,3-dimethylphenoxyl radical was the most inhibiting elementary step in the consumption of T135MB at $\Phi = 0.4$.

Reaction flux analysis for the consumption of T135MB at equivalence ratio of 0.4 (890 K) and 1.0 (950 K) with 70% conversion are presented in Fig. 6. The major consumption route of T135MB at two equivalence ratios is quite similar except for some slight differences. At two equivalence ratios, major consumption routes of T135MB are metatheses by HO_2 , OH, H and CH_3 to produce 3,5-dimethylbenzyl radicals, which are quite similar to the oxidation pathways in the flow reactor [6]. Among these reactions, metatheses by OH radical are the most dominant consumption channel for both cases. ROP analysis indicates that a large amount of HO_2 is

formed from the oxidation reaction of HCO and H under both conditions. Produced HO_2 then converts into OH through the dehydrogenation of 3,5-dimethylbenzyl. Sensitivity analysis in Fig. 5 also indicates that reaction $H + O_2 (M) = HO_2 (M)$ has a significant promoting effect to T135MB oxidation at low-temperature. However, the second dominant consumption pathway is different. Under lean condition, metatheses of T135MB with O atom are the second important consumption pathway of T135MB with a contribution of about 14%. While under stoichiometric condition, metatheses of T135MB with H atom are the second important consumption pathway of T135MB with a contribution of about 17%. According to both ROP and the sensitivity analysis, this difference between lean and stoichiometric conditions can be attributed to the higher oxygen concentration under lean condition. Minor oxidation channels of T135MB yield to m-xylene, 1,3,5-trimethylphenol and 1,3,5-trimethylphenyl radicals. Similarly, main consumption pathways of O_2 are also different between lean and stoichiometric condition. Under lean condition, the reaction of $HCO + O_2 = CO + HO_2$ dominates the consumption of O_2 , while under stoichiometric condition, most of O_2 reacts with D13MBJ to produce D13MBOO.

Under stoichiometric condition, most of 3,5-dimethylbenzyl is consumed by the combination of two 3,5-dimethylbenzyl radicals to form one 1,2-bis(3,5-dimethylphenyl) ethane. 1,2-bis(3,5-dimethylphenyl) ethane is further converted into 3,5-dimethylphenacyl radical through a concerted elimination with H atom and other radicals. Under lean condition, reaction of 3,5-dimethylbenzyl with HO_2 radical to form 3,5-dimethylbenzaldehyde becomes the dominant consumption pathway of 3,5-dimethylbenzyl, which accounts for 46% of whole consumption of 3,5-dimethylbenzyl radicals for equivalence ratios of 0.4. Corresponding value under stoichiometric condition is 24%. In the work of Diévert et al. [6], oxidation of T135MB was performed under stoichiometric condition by freely propagation flame, flow reactor and shock tube at 1, 12.5 and 20.4 atm, respectively. In these experiments at $\Phi = 1.0$, consumption route of T135MB is mainly H-abstraction reactions of T135MB but reaction of forming 1,2-bis(3,5-dimethylphenyl) ethane is no longer to be dominant reaction route for consumption of 3,5-dimethylbenzyl radical under stoichiometric condition.

ROP analysis indicates that most of 1,2-bis(3,5-dimethylphenyl) ethane and 3,5-dimethylbenzaldehyde finally convert to 1,3-dimethylbenzoyl. Role of 1,3-dimethylbenzoyl is important in the low-temperature oxidation in JSR. It is mainly formed from the reaction of 3,5-dimethylbenzaldehyde with OH and H. It is mainly consumed by decomposition to produce 3,5-dimethylphenyl and carbonyl radical through the CO elimination reaction. 3,5-Dimethylphenyl radicals are generally converted to 3,5-dimethylbenzoxo radicals with O atom, which play significant transitional roles in the formation of other intermediates during the oxidation of T135MB. Through a series of elementary steps, 3,5-dimethylbenzoxo is finally converted to light hydrocarbons such as propyne and propenyl radical.

4.3. T135MB oxidation in flow reactor

Diévert et al. [6] investigated the flow reactor oxidation of T135MB/ O_2/N_2 mixtures at temperatures of 650–970 K, under 12.5 atm and stoichiometric condition with residence time of 1.8 s. Mole fraction profiles of reactants and major species were determined using Fourier Transform Infrared (FTIR) spectrometer for water and T135MB, eletrochemical oxygen analyzer (O_2), and non-dispersive infrared analyzer for CO and CO_2 . To validate the universality of present model against experiments in flow reactor, it was used to simulate the flow reactor oxidation. The experimental and predicted mole fraction profiles by present model and Diévert et al. [6] mechanism are shown in Fig. 7. The onset of

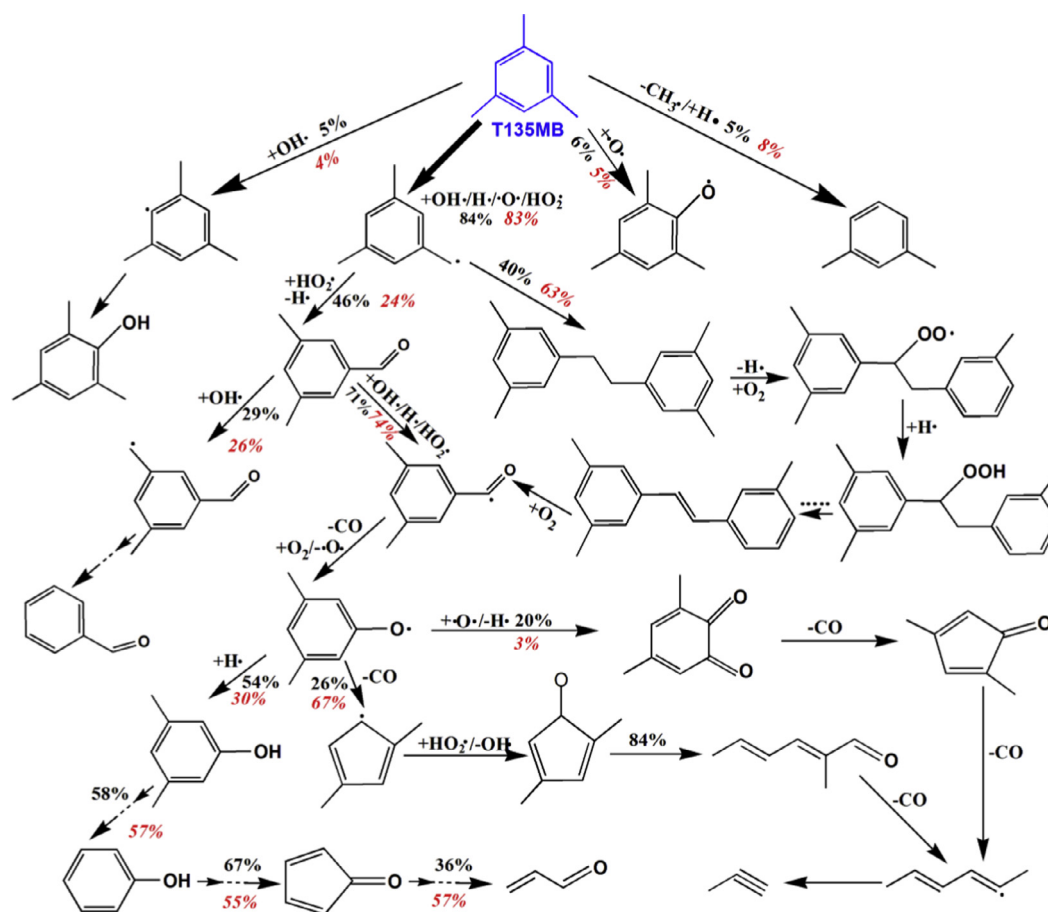


Fig. 6. Reaction flux analysis of T135MB oxidation under lean and stoichiometric (italic) conditions, corresponding to 890 K (70% conversion) and 950 K (70% conversion), respectively.

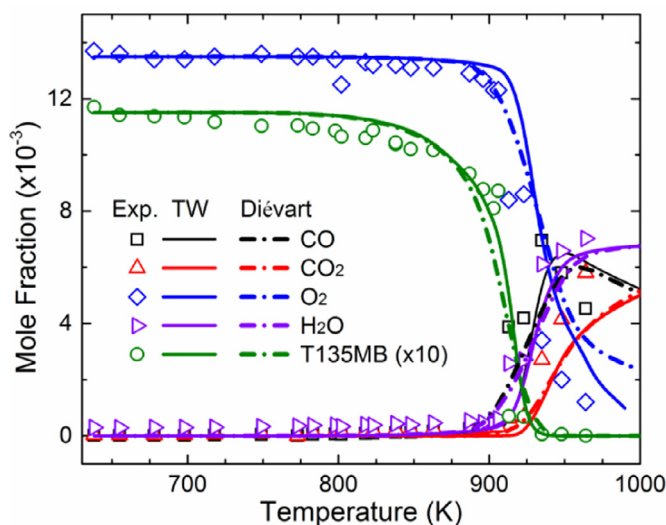


Fig. 7. Experimental and simulated mole fraction profiles of T135MB oxidation in Princeton flow reactor. Symbols are the experimental results (Exp.) from Diévert et al. [6]. Solid (TW) and dash-dotted (Diévert) lines are the simulated results with the present model and the Diévert model [14], respectively.

T135MB consumption is observed to occur at 930 K in flow reactor, which is higher than that of T135MB oxidation in JSR. This discrepancy may indicate that T135MB has less reactivity in flow reactor than in JSR. This difference can be attributed to the longer

residence time in JSR, which promote the contribution of H-atom abstraction of T135MB by OH radical. In general, present mechanism and Diévert's model can both predict well the experimental trends of T135MB, O₂, CO, CO and H₂O. It should be noted that present model exhibits a better agreement with measured data of T135MB and O₂ than those predicted by Diévert's model.

4.4. Global T135MB combustion behavior: ignition delay times

There are many previous experimental studies on the measurements of ignition delay times of T135MB, which covers wide ranges of temperatures, pressures and equivalence ratios. In the present work, the ignition delay times of T135MB measured by Diévert et al. [6] and Rao et al. [10] are used to validate the present model. The simulations are performed using constant volume closed homogeneous bath reactor in the Chemkin-II software [28].

In a shock tube study [6], ignition delay times of T135MB/air mixtures were measured in the temperature range of 1111–1468 K, at pressure of 10 and 20 atm and equivalence ratios of 0.5, 1.0 and 2.0. The time interval between the shock arrival/reflection at the end wall and onset of ignition as apparent by measurement of electronically-excited OH chemiluminescence observed at the end wall is defined as the ignition delay times. Ignition delay times of T135MB are predicted by present model under these experimental conditions. Figure 8 presents the experimental data from Diévert et al. [6] and simulated results by current mechanism. The previous model proposed by Diévert et al. [6] under-predicts the fuel reactivity under rich-fuel conditions with the largest experimen-

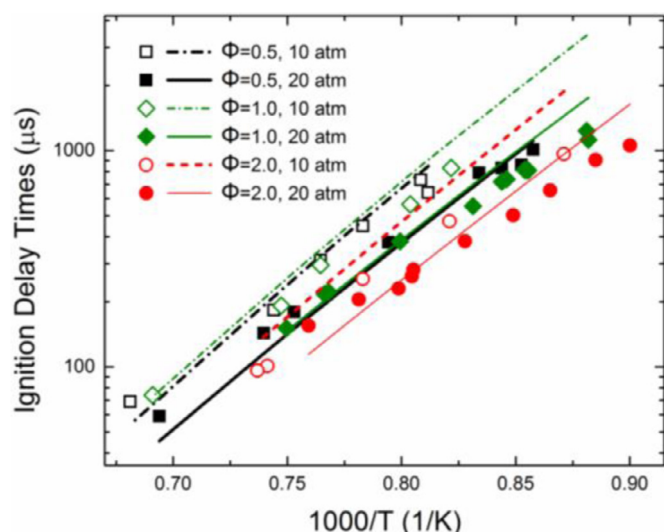


Fig. 8. Experimental and simulated reflected shock ignition delay time for T135MB/air mixture. Symbols are the experimental results from Diévert et al. [6], and lines are the simulated results by the present model.

tal modeling difference a factor of 1.5. Though there is an over-prediction of the ignition delay times at 10 atm, present model tends to predict well the data measured at 20 atm from lean to rich conditions, indicating that current model is generally valid in moderate temperature and wide equivalence ratio as well as pressure conditions.

Rao et al. [10] measured the ignition delay times of T135MB mixture behind reflected shock waves in a heated shock tube. The experiments were performed over temperature range of 1080–1560 K, pressures of 1.0–5.0 atm, and equivalence ratios of 0.5, 1.0 and 2.0. The time interval between the steep rise of pressure and the maximum rate of increase in the CH^* emission signal at the side wall is defined as the ignition delay time. Figure 9 presents the experimental data from Rao et al. [10] and the modeling results by the present model. It can be seen that present model can reproduce the ignition delay times of T135MB from fuel-lean to fuel-rich conditions. Current models have a better agreement with experimental data at pressure of 1.0 and 5.0 atm than the simulated results by Rao et al. [10] who used the Gudiya mechanism, which is developed mainly for the simulation of T135MB oxidation at high pressures [21].

5. Conclusion

Low temperature oxidation of 1,3,5-trimethylbenzene in a JSR was investigated over temperature range from 700 to 1100 K at atmospheric pressure. Mole fraction profiles about 22 species including reactants, stable intermediates and products were measured by online GC techniques under lean and stoichiometric conditions. In general, low-temperature oxidation of T135MB at stoichiometric condition tends to shift toward higher temperature region with more light hydrocarbon intermediates than that under fuel-lean condition. Based on quantum chemistry calculation and experimental results, a detailed kinetic model of T135MB was developed to simulate the oxidation characteristic of T135MB and formation of stable intermediates in low-temperature JSR. Moreover, it was validated against the experimental data in present work. The main consumption pathway for T135MB low-temperature oxidation are H-abstraction reactions to produce 3,5-dimethylbenzyl radicals, while metatheses forming m-xylene and reactions with O/OH radicals to generate 1,3,5-trimethylphenoxy/1,3,5-trimethylphenyl play minor roles. Metatheses of T135MB with HO_2/OH have promoting effect on the consumption of T135MB at both lean and stoichiometric conditions. H-abstraction attacked by OH radical forming 3,5-dimethylbenzyl radical was the most inhibiting reaction at $\Phi = 1.0$. While hydrogenation of 1,3-dimethylphenoxy radical was the most inhibiting elementary step at $\Phi = 0.4$.

Moreover, present model was validated against other different types of experimental data on the literature, including flow reactor oxidation as well as global combustion property ignition delay times. Good performance of present model in reproducing these results exhibits its applicability under a wide range of combustion conditions. This work establishes a foundation for further investigation of surrogate fuel containing T135MB under wide experimental conditions.

Acknowledgments

This research is mostly supported by the Natural Science Foundation of China (no 91541102/51476168), Ministry of Science and Technology of China (2017YFA0402800) and the Recruitment Program of Global Youth Experts. Additional supports from the CAS Interdisciplinary Team and the Science and Technology on Liquid Rocket Engine Laboratory are also acknowledged. The authors are grateful to Prof. Li-Dong Zhang from University of Science and Technology of China for his help in the theoretical calculations.

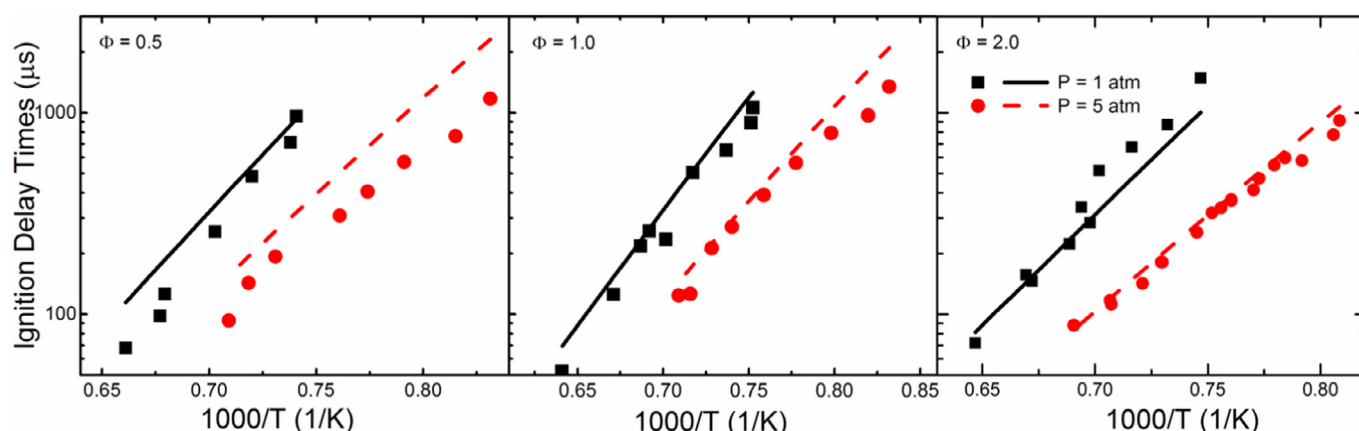


Fig. 9. Experimental and simulated reflected shock ignition delay times for T135MB/air mixture. Symbols are the experimental results from Rao et al. [10], and lines are the simulated results with the present model.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2017.12.030.

References

- [1] J.D. Munzar, B. Akih-Kumgeh, B.M. Denman, A. Zia, J.M. Bergthorson, An experimental and reduced modeling study of the laminar flame speed of jet fuel surrogate components, *Fuel* 113 (2013) 586–597.
- [2] A.G. Gasanov, A.G. Azizov, I.M. Akhmedov, I.G. Ayubov, G.D. Gasanova, Obtaining jet fuel components by decarboxylation of crude oil acids in the presence of nanosized titanium dioxide, *Chem. Technol. Fuels Oils* 49 (2013) 329–332.
- [3] T. Malewicki, S. Gudiya, K. Brezinsky, Experimental and modeling study on the oxidation of Jet A and the n-dodecane/iso-octane/n-propylbenzene/1,3,5-trimethylbenzene surrogate fuel, *Combust. Flame* 160 (2013) 17–30.
- [4] O.D. Adeniyi, A.S. Kovo, A.S. Abdulkareem, C. Chukwudozie, Ethanol fuel production from cassava as a substitute for gasoline, *J. Dispersion Sci. Technol.* 28 (2007) 501–504.
- [5] C.P. Wood, V.G. McDonnell, R.A. Smith, G.S. Samuelsen, Development and application of a surrogate distillate fuel, *J. Propul. Power* 5 (1989) 399–405.
- [6] P. Diévert, H.H. Kim, S.H. Won, Y. Ju, F.L. Dryer, S. Dooley, W. Wang, M.A. Oehlschlaeger, The combustion properties of 1,3,5-trimethylbenzene and a kinetic model, *Fuel* 109 (2013) 125–136.
- [7] A. Roubaud, R. Minetti, L.R. Sochet, Oxidation and combustion of low alkylbenzenes at high pressure: comparative reactivity and auto-ignition, *Combust. Flame* 121 (2000) 535–541.
- [8] X. Hui, A.K. Das, K. Kumar, C.-J. Sung, S. Dooley, F.L. Dryer, Laminar flame speeds and extinction stretch rates of selected aromatic hydrocarbons, *Fuel* 97 (2012) 695–702.
- [9] C. Ji, E. Dames, H. Wang, F.N. Egolfopoulos, Propagation and extinction of benzene and alkylated benzene flames, *Combust. Flame* 159 (2012) 1070–1081.
- [10] F. Rao, B. Li, P. Li, C. Zhang, X. Li, Shock-tube study of the ignition of gas-phase 1,3,5-trimethylbenzene in air, *Energy Fuels* 28 (2014) 6707–6713.
- [11] S.H. Won, S. Dooley, F.L. Dryer, Y. Ju, Kinetic effects of aromatic molecular structures on diffusion flame extinction, *Proc. Combust. Inst.* 33 (2011) 1163–1170.
- [12] C.H. Zhang, B. Li, F. Rao, P. Li, X.Y. Li, A shock tube study of the autoignition characteristics of RP-3 jet fuel, *Proc. Combust. Inst.* 35 (2015) 3151–3158.
- [13] R.K. Gehmlich, A. Kuo, K. Seshadri, Experimental investigations of the influence of pressure on critical extinction conditions of laminar nonpremixed flames burning condensed hydrocarbon fuels, jet fuels, and surrogates, *Proc. Combust. Inst.* 35 (2015) 937–943.
- [14] K. Seshadri, A. Frassoldati, A. Cuoci, T. Faravelli, U. Niemann, P. Weydert, E. Ranzi, Experimental and kinetic modeling study of combustion of JP-8, its surrogates and components in laminar premixed flows, *Combust. Theor. Model.* 15 (2011) 569–583.
- [15] S. Jahangirian, C.S. McEnally, A. Gomez, Experimental study of ethylene counterflow diffusion flames perturbed by trace amounts of jet fuel and jet fuel surrogates under incipiently sooting conditions, *Combust. Flame* 156 (2009) 1799–1809.
- [16] K. Brezinsky, The high-temperature oxidation of aromatic hydrocarbons, *Prog. Energy Combust. Sci.* 12 (1986) 1–24.
- [17] S.H. Won, S. Dooley, F.L. Dryer, Y. Ju, A radical index for the determination of the chemical kinetic contribution to diffusion flame extinction of large hydrocarbon fuels, *Combust. Flame* 159 (2012) 541–551.
- [18] S.H. Won, S. Dooley, F.L. Dryer, Y. Ju, Measurement of extinction limits and OH radicals for trimethylbenzene and n-propylbenzene diffusion flames, 48th AIAA Aerospace Sciences Meeting Including the New Horizons Forum and Aerospace Exposition (2010), p. 197.
- [19] G. Bikas, Kinetic Mechanisms for Hydrocarbon Ignition PhD thesis, RWTH Aachen University, Aachen, Germany, 2001.
- [20] Y.Y. Zhu, S.J. Li, D.F. Davidson, R.K. Hanson, Ignition delay times of conventional and alternative fuels behind reflected shock waves, *Proc. Combust. Inst.* 35 (2015) 241–248.
- [21] S. Gudiya, K. Brezinsky, High pressure study of 1,3,5-trimethylbenzene oxidation, *Combust. Flame* 159 (2012) 3264–3285.
- [22] B.Y. Wang, Y.X. Liu, J.J. Weng, P. Garborg, Z.Y. Tian, New insights in the low-temperature oxidation of acetylene, *Proc. Combust. Inst.* 36 (2017) 355–363.
- [23] J.J. Weng, Y.X. Liu, B.-Y. Wang, L.L. Xing, L.D. Zhang, Z.Y. Tian, Experimental and kinetic investigation of 1,2,4-trimethylbenzene oxidation at low temperature, *Proc. Combust. Inst.* 36 (2017) 909–917.
- [24] J.A. Montgomery, M.J. Frisch, J.W. Ochterski, G.A. Petersson, A complete basis set model chemistry. VI. Use of density functional geometries and frequencies, *J. Chem. Phys.* 110 (1999) 2822–2827.
- [25] M. Frisch, G. Trucks, H. Schlegel, G. Scuseria, M. Robb, J. Cheeseman, G. Scalmani, V. Barone, B. Mennucci, G. Petersson, Gaussian 09, Revision D. 01, Gaussian, Inc., Wallingford CT, 2009.
- [26] C. Gonzalez, H.B. Schlegel, An improved algorithm for reaction path following, *J. Chem. Phys.* 90 (1989) 2154–2161.
- [27] V. Mokrushin, V. Bedanov, W. Tsang, M. Zachariah, V. Knyazev, ChemRate, Version 1.19, NIST, Gaithersburg, MD, 2002.
- [28] R.J. Kee, F.M. Rupley, J.A. Miller, Sandia Report: SAND 89-8009B (1989).
- [29] M.A. Oehlschlaeger, D.F. Davidson, R.K. Hanson, High-temperature thermal decomposition of isobutane and n-butane behind shock waves, *J. Phys. Chem. A* 108 (2004) 4247–4253.
- [30] Z.Y. Tian, W.J. Pitz, R. Fournet, P.-A. Glaude, F. Battin-Leclerc, A detailed kinetic modeling study of toluene oxidation in a premixed laminar flame, *Proc. Combust. Inst.* 33 (2011) 233–241.
- [31] G. da Silva, M.R. Hamdan, J.W. Bozzelli, Oxidation of the benzyl radical: mechanism, thermochemistry, and kinetics for the reactions of benzyl hydroperoxide, *J. Chem. Theor. Comput.* 5 (2009) 3185–3194.
- [32] W.H. Yuan, Y.Y. Li, P. Dagaut, J.Z. Yang, F. Qi, Investigation on the pyrolysis and oxidation of toluene over a wide range conditions. I. Flow reactor pyrolysis and jet stirred reactor oxidation, *Combust. Flame* 162 (2015) 3–21.
- [33] M. Frenklach, Reaction mechanism of soot formation in flames, *Phys. Chem. Chem. Phys.* 4 (2002) 2028–2037.